

Research papers

Optimal microgrid operation considering accurate battery degradation—World model-based reinforcement learning approach[☆]

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ABSTRACT

Modern microgrids incorporate self-generating energy devices like Photovoltaic (PV) systems along with Energy Storage Systems (ESS). Effective scheduling of energy demand, generation, and storage in these systems is crucial for economic efficiency. However, optimizing a microgrid is challenging due to its complexity and uncertainties. Reinforcement learning (RL) has proven effective in making robust, sequential decisions under the uncertainties inherent in microgrid optimization. However, many RL methods frequently fail to take advantage of utilizing future information in decision-making. This study proposes an World Model-based RL under the World Model framework that leverages prediction models for future PV generation and power demand, enabling the RL agent to make more informed and effective decisions in PV-ESS operations. It is essential to consider battery degradation expenses to ensure cost-effective microgrid operation. Although ESS can lower electricity costs by storing and shifting energy, managing the trade-off with degradation costs is crucial in the optimization process. Our approach utilized an accurate degradation model suitable for RL, ensuring a comprehensive cost objective. Simulations conducted within a university campus setting showed notable cost reductions ranging from 0.3% to 25.3% compared to benchmark models.

1. Introduction

In recent years, renewable energy has gained global attention for its environmental benefits in reducing carbon emissions and its potential to lower electricity costs [1,2]. These benefits sparked interest, especially among high-energy-consuming facilities, such as large buildings, factories, universities, and so on [3]. In response to this rising interest, the PV-ESS microgrid system, which combines photovoltaic (PV) renewable generation and energy storage systems (ESS), has emerged. This system can offer economic benefits by reducing electricity costs through peak shaving and enabling the use of cheaper electricity during high-cost periods under time-of-use (ToU) pricing (see Table 1).

However, the time-varying and nonlinear nature of the PV-ESS system poses challenges in achieving actual cost reductions [1,4]. The time-varying nature of the PV-ESS system stems from two key factors. First, PV generation fluctuates with solar irradiance changes throughout the day, resulting in variable output. Second, power demand exhibits temporal variations, influenced by factors such as irregular demand patterns based on the facility's operational hours and electricity pricing [5,6]. The nonlinear behavior of batteries is apparent

in their performance, as degradation occurs over time in a non-linear manner influenced by State of Charge (SoC), State of Health (SoH), and external environmental conditions [7].

These factors significantly impact PV-ESS system performance, but many existing studies tend to oversimplify this complexity. For example, the study of [8] used a fixed demand profile, while [9] linearized the power demand model to solve the optimization problem. Furthermore, studies by [10,11] assumed a linearized degradation model for the nonlinear degradation process, and the study of [12] focused only on the number of life cycles to describe battery degradation, neglecting other important factors.

To optimize the economic performance of PV-ESS systems, it is essential to address three key elements:

1. **Accurate prediction models** for demand and PV generation that account for time-varying characteristics.
2. **Battery degradation models** that capture the nonlinear nature of battery performance.

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Table 1
Nomenclature.

Notation	Definition
O_t	Observation at time t
S_t	State at time t
A_t	Action at time t
R_t	Reward at time t
R_t^{elec}, R_t^{deg}	Electricity purchase cost, degradation cost at time t
h_t	Hour of the day at time t
X_t	Battery SoC at time t
$\bar{X}_t$	Maximum battery capacity at time t
Z_{PV}, Z_D	PV and Demand predicted latent value
D_t	Electricity demand between time t and $t+1$
G_t	Electricity generated amount from PV between time t and $t+1$
P_t	Electricity purchase amount from supplier between time t and $t+1$

3. Effective optimization methods designed to handle uncertainties in PV-ESS systems.

Table 2 summarizes the previous efforts according to the above key elements. In this study, we propose a World Model based Reinforcement Learning (RL) framework that incorporates prediction models. In addition to prediction models, we also considered battery degradation to facilitate more effective decision-making in the operation of PV-ESS systems.

1. **Predictions of future behaviors** can enhance the optimization process in systems with such time-varying uncertainties. Previous research has proposed various approaches for accurately predicting PV generation, including statistical time series analysis models, as well as machine learning (ML) and artificial intelligence (AI) based models. Statistical models analyze patterns in past PV generation data to predict future values. For example, the study of [13] used statistical models such as ARIMA to predict PV generation in various regions. These methodologies have predictive performance limitations because they overlook exogenous variables such as solar irradiance and temperature, which are critical as they directly influence PV generation [14].

Recent studies employ AI and ML-based models that take these variables into account, particularly using neural network models such as recurrent neural networks (RNN) and long short-term memory (LSTM) networks to enhance the accuracy of PV generation predictions by considering the temporal continuity of time series data [15–19]. Ensemble techniques reduce uncertainty and improve prediction accuracy by combining the results of multiple prediction models [20,21]. Moreover, attention mechanisms are employed to enhance prediction accuracy by assigning more importance to crucial information [22].

However, most AI/ML-based PV prediction models have not fully utilized weather data, which can be divided into forecast data (future weather predictions) and observation data (historical records). For instance, studies by [16,17] developed prediction models using solely PV data without integrating weather information, while [20] relied exclusively on weather forecast data. Similarly, the research by [21,22] utilized only weather observation data. In contrast, studies utilizing observation and forecast data demonstrate higher prediction accuracy. For example, the authors of [15,23] achieved superior prediction performance by combining both data compared to using a single data source. Yet, these studies employed relatively older models, such as simple RNNs or multi-layer perceptrons, indicating the need for more advanced models. To overcome these limitations, we propose a PV prediction model that integrates an attention mechanism with a Gated Recurrent Unit (GRU). Our model comprehensively uses both weather observation and forecast data, employing a more advanced model.

In addition to PV generation prediction, accurate power demand forecasting is also essential. While the main objective of PV-ESS systems is to provide stable and economical power [24], large institutions with irregular demand patterns can find it challenging to achieve this goal, as such variability complicates demand predictions [25]. For these entities, accurate demand prediction models are crucial as they can

help minimize uncertainty. To develop an accurate demand prediction model, this study uses cyclical encoding techniques during data preprocessing to reflect the periodicity of time information. Periodic features at various time scales, such as months, days, and hours, are integrated into the model. Additionally, a GRU-based neural network model capable of learning long-term dependencies in sequence data is employed.

2. **Battery degradation** is another crucial factor to consider in PV-ESS system optimization. While ESS can offer economic advantages, they represent a significant investment, and their lifespan decreases with usage. Therefore, accurately calculating the degradation cost is essential for PV-ESS operation. To accurately reflect the non-linear nature of degradation, considering both SoC and Depth of Discharge (DoD) is necessary [26]. However, current studies on ESS operation have either employed simplified models or overlooked ESS degradation. For instance, the authors of [27] utilized the maximum value between the SoC and DoD costs. The study by [10] considered both DoD and SoC but linearized the degradation model. This approach fails to accurately reflect the non-linearity. Studies by [28,29] only considered single factors such as DoD or SoC. Some studies by [30,31] even ignored degradation during operation. In this paper, we employ the Markovian degradation model proposed by [32], which considers both DoD and SoC factors and is well-suited for an RL framework. The Markovian model builds upon the high-performing degradation model developed by [26], while overcoming its limitation of requiring full operational history for degradation calculations. Therefore, the Markovian model adheres to the Markov property, as it uses the current state without relying on past data, making it suitable for RL applications.

3. **Various control methodologies** have been employed to optimize PV-ESS operations, as PV-ESS microgrids have various interacting components. Therefore, implementing effective methodologies to optimize these complex operations is crucial. For example, the authors of [33] optimized the operation of grid-connected wind and solar microgrid by employing the Model Predictive Control (MPC) algorithm. Authors of [34] presented battery control of the New York region using the MPC algorithm. While [35] proposed a tube-based MPC to enhance robustness against forecast uncertainties in microgrid operation. The study by [36] performed microgrid optimization using the mixed-integer linear programming (MILP) technique. These studies primarily utilized mathematical optimization approaches for effective operation. However, these approaches require precise future information and struggle with uncertainties, as the solutions can change when the environment shifts. On the other hand, RL has proven to be well-suited for sequential decision-making under uncertainties. RL can learn the environmental dynamics without needing accurate future information. For instance, the authors of [37] used Q-learning to optimize microgrid with ESS and gas turbine. The study of [38] used RL for optimal operation with flexible loads. Other studies, such as those by [39], proposed a multi-agent RL method for microgrid operation, while [30] developed an arbitrage strategy to maximize operating profit using RL. Table 2 presents a comparative analysis of previous microgrid operational literature and our proposed method.

Despite the strengths of RL in uncertain environment, most RL-based microgrid studies often overlook the use of predicted future values. Incorporating highly accurate predictions can help manage uncertainties and improve learning effectiveness. Therefore, this study employs a World Model-based RL framework, which uses prediction models to simulate future dynamics to effectively train the RL agent [40]. This approach allows for robust policy formation under uncertain environmental conditions.

To summarize, existing approaches reveal three major limitations. First, classic optimization methods such as LP and QP assume perfect foresight, making them impractical for real-time operation. MPC, although applicable in real-time, still relies heavily on forecast accuracy. In practice, having perfectly accurate future information is nearly impossible. Second, while RL-based approaches are well-suited

Table 2
Comparison of Microgrid Operation Papers.

Paper	Predicted Future Information	Battery Degradation	Control Methodologies	Utilized Actual Data
[31]	N/A	N/A	Genetic algorithm	–
[38]	N/A	N/A	RL(A3C, PPO)	–
[39]	N/A	N/A	RL(MADQN)	✓
[29]	N/A	SoC	MINLP	✓
[33]	N/A	DoD	MPC	✓
[32]	N/A	SoC and DoD	RL (DQN, SAC)	–
[18]	PV, Demand	N/A	LP	–
[19]	PV	N/A	DP	✓
[36]	PV, Demand	N/A	MILP	✓
[30]	Market Price, Demand	N/A	RL(Q-learning)	✓
[37]	PV, Demand	N/A	RL(Q-learning)	✓
[35]	PV, Demand	DoD	MPC	✓
[34]	PV	SoC and DoD	MPC	✓
[27]	PV, Wind	SoC or DoD	Genetic algorithm	✓
[10]	Market Price	SoC and DoD (Linearized)	RL(NN-DDQN)	✓
Proposed method	PV, Demand	DoD and SoC	RL(WMDDPG)	✓

for uncertainty, they rarely leverage forecasted information. Third, most degradation models are either linearized or simplified, considering only a single factor such as SoC or DoD. Few studies have attempted to integrate prediction, robust control, and precise degradation simultaneously in a unified framework.

Table 2 compares according to the four aspects: predicted future information, battery degradation, control methodologies, and actual data usage denoting whether real operational data from microgrids was utilized rather than assumptions. Papers incorporating predictions mainly focused on forecasting renewable energy sources, demand, and market prices. Battery degradation was not considered in many studies; those that did account for it typically used single factors or employed simplified models. The control methodologies exhibit diverse approaches, from conventional methods to RL techniques. While papers typically addressed either prediction mechanisms or battery degradation independently, few studies attempted to integrate both aspects comprehensively. Our proposed method simultaneously incorporates both PV and demand predictions while considering complete battery degradation factors within a World Model RL framework.

The objective of this study is to apply a World Model-based RL model to the PV-ESS system and propose an optimal operation strategy. We aim to reduce system variable uncertainty through prediction models and establish cost-minimizing policies considering battery degradation. To validate our model's effectiveness, we simulated our model with a real-world microgrid environment using campus data.

The contributions of this study are summarized as follows:

1. We propose a World Model-based RL framework incorporating high-performance PV generation and power demand prediction models to effectively address the uncertainty factors in PV-ESS systems.
2. We integrate a battery degradation model into our framework, enabling more realistic and cost-effective operational strategies by considering degradation costs in economic operation.
3. Our proposed model demonstrates performance closest to the optimal solution among most of benchmark methods and achieves cost savings of 0.3 - 25.3% across various test cases.

The remainder of this paper is structured as follows. Section 2 covers the preliminaries, providing background knowledge. Section 3 offers an overview of the system architecture and its components. Section 4 elaborates on the methods, detailing the implementation of the prediction models, RL, and the World Model. Section 5 presents the results, including the setup and outcomes. Finally, Section 6 concludes the study.

2. Background

2.1. World model

The World Model is an RL approach that determines the optimal action utilizing the predicted future. It consists of a vision model (V), a memory model (M), and a controller model (C) [40]. Fig. 1 illustrates the operation of the World Model. The vision model compresses the observed information into a latent vector, the memory model predicts the next state based on the latent vector and action, and the controller model utilizes the observed values and future information to derive the optimal action.

$$a_t = W_c[z_t, h_t] + b_c, \quad (1)$$

where z_t represents the currently observed information generated through the encoder of the Variational Autoencoder (VAE), and h_t represents the predicted future information generated through the memory model. In Eq. (1), W_c and b_c are the weight matrix and bias of the controller model, respectively. Z_t is latent information from V and h_t is predicted value from M (i.e., the output of prediction model). Lastly, a_t is an action value from C (i.e., the RL agent's decision). The World Model combines the observed and the predicted information to estimate actions through a linear model, enabling more effective strategies to be derived by leveraging future information.

2.2. Deep reinforcement learning

RL is a method that allows an agent to discover a policy that maximizes future cumulative reward (i.e., $\sum_{t=0}^{\infty} \gamma^t R_{t+i}$) in an environment defined by the Markov Decision Process (MDP) [41]. Deep Reinforcement Learning (DRL) employs deep neural networks to approximate value functions or policies, enabling effective learning in complex, high-dimensional environments [42]. The action-value function $Q(s, a)$ represents the expected cumulative reward of taking action a in state s .

$$Q(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a] \quad (2)$$

The optimal action a^* to take in the current state can be chosen as the action with the highest Q-value within the possible action set A :

$$a^* = \operatorname{argmax}_{a \in A} Q(s, a) \quad (3)$$

Q-learning, one of the model-free RL algorithms, learns the optimal action-value function directly without needing to know the environment's transition probabilities. The update rule for Q-learning is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a)], \quad (4)$$

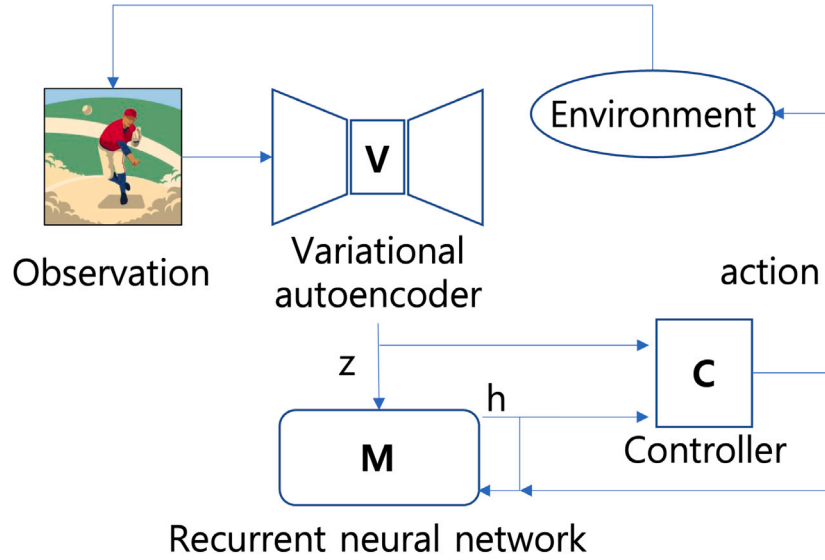


Fig. 1. World Model architecture.

Source: Figure adapted from [40].

where α is the learning rate, r is the reward received after taking action a , γ is the discount factor, and s' is the next state.

2.3. ESS degradation model

Batteries experience life degradation, where the available capacity decreases with usage. Although battery degradation may be influenced by weather factors such as temperature and humidity, degradation is mainly affected by its usage patterns. Many studies have been conducted on battery degradation, and [26] proposed a mathematical model that considers both SoC and DoD based on battery usage patterns.

$$\bar{L} = \alpha e^{-N\beta f_{d,1}} + (1 - \alpha)e^{-Nf_{d,1}} \quad (5)$$

Equation (5) provides a method for estimating the lithium-ion battery SoH, noted as $\bar{L}$. The $\bar{L}$ value ranges between 0 and 1, where a value of 1 represents a new battery with full capacity, and a value of 0 represents a battery that depleted its capacity. In this equation, the parameters α and β are factors that contribute to battery degradation, and N represents the total number of accumulated cycles. Additionally, $f_{d,1}$ indicates the degradation that occurs in a lithium-ion battery after one cycle, as influenced by the battery's usage pattern. Although the degradation model of [26] shows excellent performance, this assumes regular charge and discharge patterns, which is unrealistic for battery operations. In the case of irregular usage patterns, the full battery operation history is required, making it dependent on past information. Since decisions in RL should be based solely on the current state, this dependency makes it difficult to apply in the RL framework. To overcome these limitations, the study of [32] proposed a Markovian degradation model based on [26].

$$|\bar{L}_t - \bar{L}_{t+1}| = f_{deg}(X_t, X_{t+1}, L_t) \quad (6)$$

The Markovian model takes the current SoC (X_t) and the next SoC (X_{t+1}) as inputs and estimates the number of cycles (N) required to reach the current SoH ($\bar{L}_t$) using the given usage pattern (X_t to X_{t+1}). Then, the Markovian model calculates the next SoH ($\bar{L}_{t+1}$) using Equation (5) and returns the degradation amount, which is the difference between the $\bar{L}_t$ and $\bar{L}_{t+1}$.

3. System architecture

This section provides an explanation of the system's main components: PV system, ESS, electricity consumer, and electricity supplier.

Fig. 2 demonstrates our system formulation, which is based on the microgrid structure of the Seoul National University of Science and Technology (SeoulTech) campus. We constructed an architecture with an RL agent that can intelligently control the PV-ESS system. Our proposed system comprises a 300 kW PV and a 500 kWh ESS. Detailed experimental settings will be discussed in Section 5.

The system comprises five key components:

1. PV: PV is a system that generates electricity using solar energy. PV generation occurs primarily during the daytime and is significantly affected by weather conditions, making it essential to account for these uncertainties when making optimal decisions. The electricity produced by the PV is stored in the ESS.

2. ESS: The ESS can be charged with power from the PV or the Korea Electric Power Corporation (KEPCO), the main energy supplier in South Korea. While the ESS enables the storage and utilization of PV-generated electricity during the daytime and electricity purchased from KEPCO, it is important to consider the degradation costs associated with its usage.

3. Power Consumer (SeoulTech): SeoulTech,¹ the electricity consumer, meet its power demand either directly from the ESS or by purchasing it from KEPCO. The power demand exhibits different patterns depending on the time of day and the season, with variations in consumption behavior throughout the year.

4. Power Supplier (KEPCO): KEPCO provides various pricing plans for different types of consumers, such as residential, industrial, and educational. Time-of-use (ToU) pricing is a system that applies different rates depending on the time of day. SeoulTech adopts the A2 W Plan, which is a ToU pricing plan designed for educational institutions.

5. RL Agent: The RL agent proposes hourly battery operations that can reduce costs by considering PV generation, power demand, hourly prices, and battery state. For generation and demand data, the agent utilizes information from prediction models. The predicted information is compressed into latent representations, providing the RL agent with condensed and informative data.

¹ Seoul National University of Science and Technology (SeoulTech) is located in Nowon-gu, Seoul, South Korea. It has an approximate annual electricity consumption of 22 GWh.

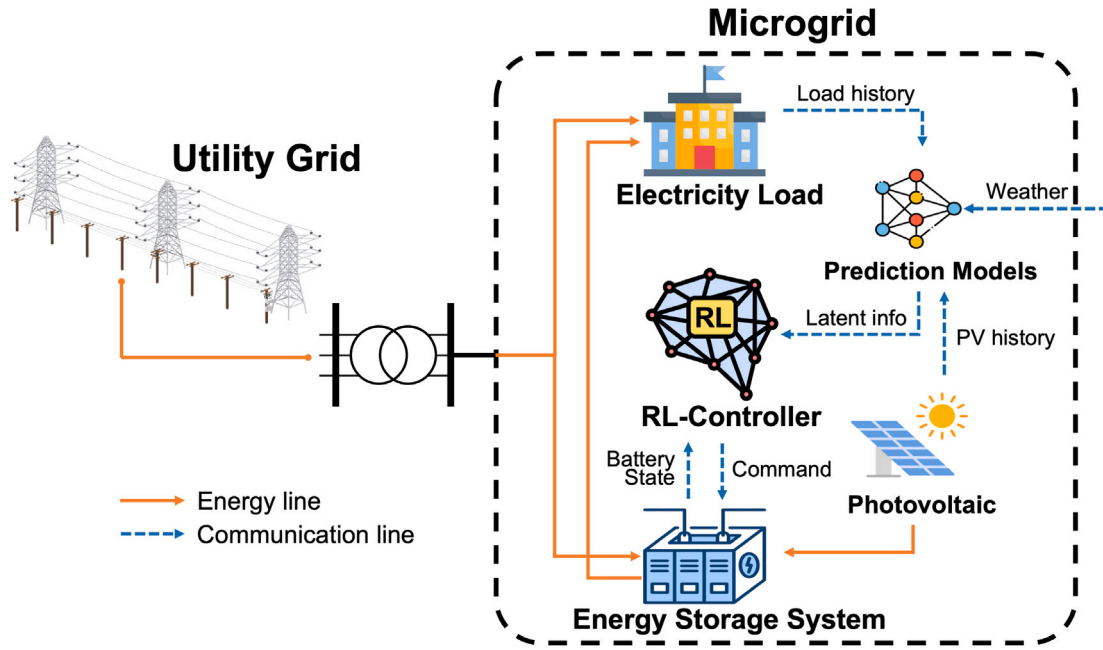


Fig. 2. Proposed architecture of SeoulTech's PV-ESS Microgrid.

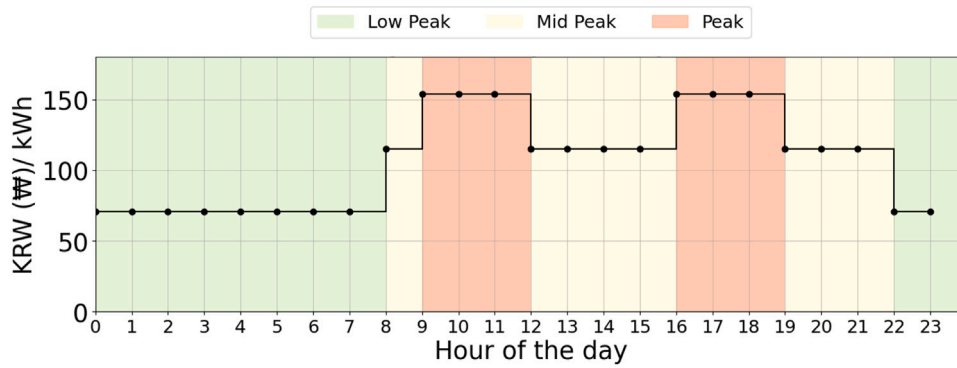


Fig. 3. Hourly electricity rates by hour [43].

3.1. Power supplier (KEPCO) pricing

KEPCO provides various pricing plans for different types of consumers, such as residential, industrial, and educational. The ToU pricing is a system that applies different rates depending on the time of day. SeoulTech adopts the A2 W Plan, which is an educational ToU pricing plan [43].

Fig. 3 shows the details of the A2 W Plan used by SeoulTech. The electricity rates are divided into three periods: Low Peak Hour (70.6 KRW/kWh), Mid Peak Hour (115.2 KRW/kWh), and Peak Hour (153.9 KRW/kWh). The A2 W Plan applies Mid Peak Hour and Peak Hour rates during the daytime when power demand is high and Low Peak Hour rates during the night when power demand is relatively low. As the difference between the lowest and highest rates is more than double, it is possible to save on electricity costs by shifting power from low-cost periods to high-cost periods using batteries.

4. Methods

4.1. PV prediction model

Accurate prediction of PV generation is crucial for optimal PV-ESS operation, and it is highly dependent on weather conditions. For

training the model, we utilized both weather observation and forecast data along with PV generation data. The weather observation and forecast data, covering the winter season from January to March, were obtained from the Korea Meteorological Administration's Open Data Portal [44]. We used data from Suwon, South Korea, the location closest to SeoulTech, where both weather forecast and observation data were accessible. The input data used for observations include temperature, humidity, sunshine duration, and past PV generation, while the forecast data include precipitation amount, rainfall type, and sky type. Temperature, humidity, sunshine duration, PV generation, and precipitation amount are continuous variables, whereas others are categorical variables. The PV generation data is used as the target variable. Fig. 4 illustrates the hourly average PV generation data. The PV generation occurs between 7 AM and 6 PM, corresponding to the daylight hours. The peak generation is concentrated between 11 AM and 2 PM when the solar irradiance is at its highest.

To achieve high prediction accuracy, we propose a model that can consider both weather observation data and weather forecast data. The architecture of our proposed prediction model is inspired by the study of [15], where the authors introduced PVHybNet, a model that predicts future PV generation using both types of weather data. PVHybNet consists of an LSTM and a Deep Feedforward Neural Network (DFNN) model, with each model using weather observation data and

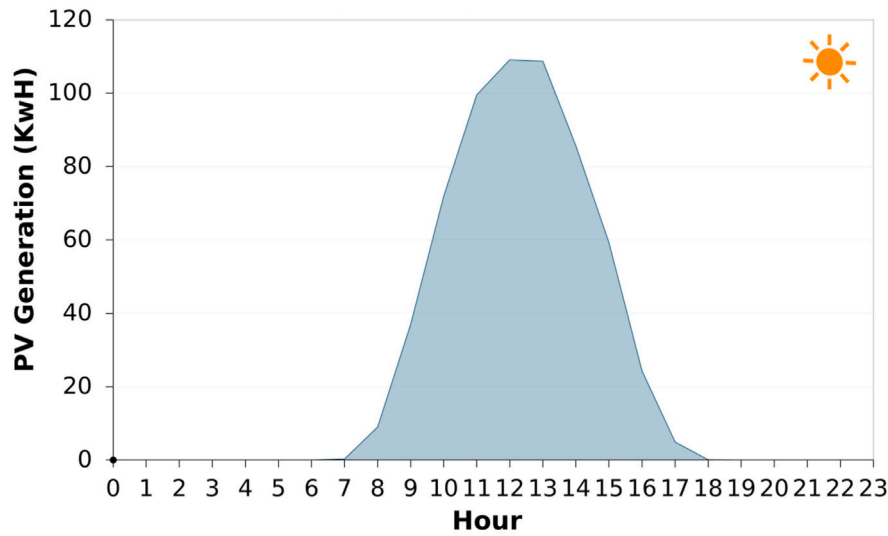


Fig. 4. Hourly average power demand of SeoulTech.

Table 3

Prediction model configuration.

Component	Value
Train-Test Split	80% training data, 20% test data
Learning Rate	0.001
Epochs	256 with early stopping
Batch Size	64
Dropout Rate	0.3
Activation Function	tanh
Optimizer	Adam

forecast data as its respective inputs. The outputs from each model are combined to predict future PV generation.

$$y_{t+n} = f(x_{t-5}^{obs}, \dots, x_t^{obs}) + g(x_{t+n}^{fst}) \quad (7)$$

Building on this framework, our proposed model introduces two modifications. First, we applied an attention mechanism to the recurrent part, enabling the model to selectively focus on the important parts of the historical observation data and extract useful features. This enhances long-term prediction accuracy by prioritizing temporal features [45]. Second, we replaced LSTM into GRU, which offers similar performance with fewer parameters and lower computational cost, making it suitable for real-time prediction [46]. The GRU model utilizes the past five hours of weather and PV data, along with weather forecasts for the next n hours, to predict n hours of future PV generation. The GRU ($f(\cdot)$ in Eq. (7)) processes observed weather and PV data from the past five hours, while the DFN ($g(\cdot)$ in Eq. (7)) handles forecasted weather data for the next n hours.

For the neural network structure, the GRU layer uses 64 units, followed by dense layers with 32, 16, and 8 units. DFN layers use 64, 32, 16, and 8 units. The outputs of the GRU and DFN are concatenated and passed through an additional dense layer to produce the final prediction. Another detailed training configuration for the PV model is described in Table 3.

4.2. Demand prediction model

Accurate forecasting of electricity demand is another crucial factor for microgrid operation. Electricity demand is influenced by various factors, such as seasonality, time of day, and usage patterns, making it a challenging task. In this study, we utilized the 2022 electricity demand data from SeoulTech, which covers the winter season from January to March. We developed an accurate electricity demand prediction

model using the advanced time series forecasting algorithm GRU. As demonstrated in the PV prediction model, GRU provides a light and accurate alternative, which makes it suitable for integration within the microgrid operation framework.

Fig. 5 presents the hourly average electricity demand at SeoulTech. The electricity demand is relatively higher during the daytime hours when students and faculty members are on campus. A slight decrease in electricity demand is observed during the lunch hour (12 PM). While there are general patterns in electricity demand related to the time of day, it can vary significantly due to various factors. The detailed forecasting results and benchmark comparisons for both PV and demand model are presented in Section 5.2.

In this study, we employed the GRU model to forecast the school's electricity demand. The electricity consumption patterns from previous steps are crucial for predicting the electricity demand in subsequent steps. Therefore, the input variables for the model include the current month, hour, and electricity consumption over the previous five hours. The structure of the GRU model mirrors that of the PV model, utilizing 64, 32, 16, and 8 units for each layer. The model's hyperparameters are identical, as detailed in Table 3.

4.3. Markov decision process

An MDP is a stochastic process satisfying the Markov property, where the transition dynamics depend only on the current state. It consists of four elements: state space (S), action space (A), reward function (R), and transition probability function (P). In this section, we introduce these four components of our MDP formulation.

4.3.1. State

The state vector (S_t) at time t consists of five elements: (1) hour (h_t), (2) state of charge (X_t), (3) maximum capacity of ESS ($\bar{X}_t$), (4) predicted electricity demand (Z_t), and (5) predicted PV generation (Z_{PV}). The predicted values (Z_t , Z_{PV}) are latent variables obtained through embedding. If the RL model uses 5-hour future predictions, the embedded values with a 2-hour horizon are used. In this case, the state size becomes 7, as it includes 2 predicted values for each prediction model. The embedding size for different future prediction horizons is discussed in detail in Section 5.

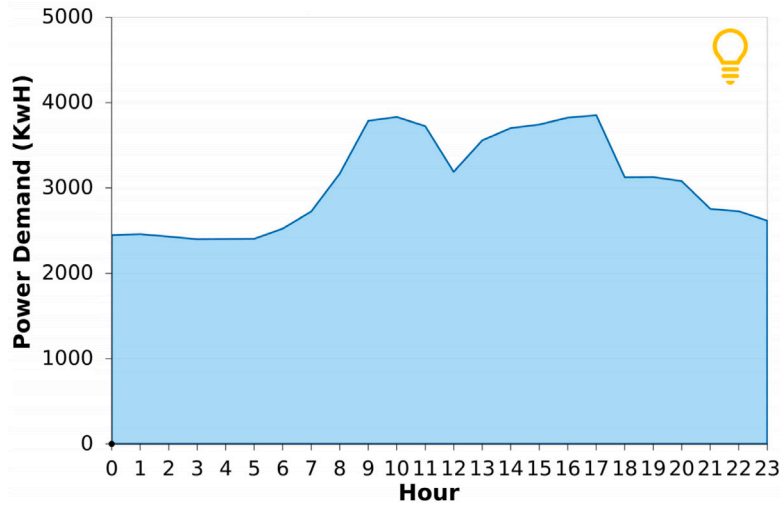


Fig. 5. Hourly average power demand.

4.3.2. Action

The action (A_t) determines the amount of electricity purchased (P_t), represented by a real number between 0 and 1. An action of 0 minimizes the electricity purchase, which means maximizing the use of the battery. An action of 1 maximizes the electricity purchase, fully charging the battery and purchasing all the demanded electricity from the supplier.

The minimum purchase is the amount of electricity that must be bought from the supplier when the battery's X_t cannot meet the demand D_t . The maximum purchase fully charges the battery to $\tilde{X}_t$ and satisfies the demand D_t by purchasing from the supplier. The PV generation G_t is assumed to be entirely available at the given time step.

$$\text{Minimum_Purchase} = D_t - X_t + G_t \quad (8)$$

$$\text{Maximum_Purchase} = \tilde{X}_t - X_t - G_t - D_t \quad (9)$$

The actual electricity purchase P_t is calculated as follows:

$$P_t = \text{Minimum_Purchase} + A_t \times (\text{Maximum_Purchase} - \text{Minimum_Purchase}) \quad (10)$$

The control interval is set to one hour, meaning the RL agent determines a new action at every hourly time step.

4.3.3. Reward

The reward is the value that an agent obtains for taking specific actions in a given state. In this research, the rewards represent the cost generated by the university. The cost comprises two main components: the purchase cost of electricity from the electricity supplier (R_t^{elec}) and the degradation of the ESS from its operation (R_t^{deg}). As the goal of RL is to maximize the total reward, the cost is defined as a negative reward.

$$\begin{aligned} R_t &= R_t^{\text{elec}} + R_t^{\text{deg}} \\ &= c_h \cdot p_t + f_{\text{deg}}(X_t, X_{t+1}, \tilde{L}_t) \cdot \text{ESS}_{\text{price}} \end{aligned} \quad (11)$$

In Eq. (11), X_{t+1} is derived from X_t and A_t , while $\tilde{L}_t$ is calculated as the ratio of $\tilde{X}_t$ to $\tilde{X}_0$. The degradation term $f_{\text{deg}}(\cdot)$ is evaluated using a Markovian model [32], which computes the battery life loss based on the current operational transition. The Markovian model quantifies the amount of battery degradation based on current SoC (X_t), next SoC (X_{t+1}), and current SoH ($\tilde{L}_t$). Specifically, the Markovian model $f_{\text{deg}}(\cdot)$ calculates battery degradation using the following procedure:

1. **Input:** Receives X_t , X_{t+1} , and $\tilde{L}_t$.

2. **Estimate N :** Using Newton's method, calculates how many cycles N needs to be repeated to reach the life $\tilde{L}_t$ under the usage pattern (X_t and X_{t+1}).
3. **Calculate $\tilde{L}_{t+1}$:** Substitutes $N + 1$ into Eq. (5) to compute $\tilde{L}_{t+1}$, which shows the battery life after one additional cycle.
4. **Return:** The $f_{\text{deg}}(\cdot)$ returns the degraded amount which is the difference between $\tilde{L}_t$ and $\tilde{L}_{t+1}$.

4.3.4. Transition

Various factors, including PV generation, power demand, and battery degradation influence the state transition. The transitions are modeled as follows:

$$X_{t+1} = X_t + P_t + G_t - D_t \quad (12)$$

$$\tilde{X}_{t+1} = \tilde{X}_t - f_{\text{deg}}(S_t, a_t, L_t) \quad (13)$$

Here, X_{t+1} is calculated by adding the purchased power (P_t) and PV generated power (G_t) and subtracting the power demand (D_t). The second equation, $\tilde{X}_{t+1}$, incorporates the battery degradation effect. It subtracts the degraded amount from the previous $\tilde{X}_t$.

4.4. World model-based reinforcement learning

Our RL agent is trained under the World Model framework, utilizing PV prediction and demand prediction models. These two prediction models forecast future information, and the predicted values are embedded into latent states. The latent states are used as part of the RL agent's state. Within the context of the World Model paradigm, the prediction models correspond to the memory part, the embedding process corresponds to the vision part, and the RL agent corresponds to the controller.

Fig. 6 illustrates the flow of our World Model-based RL. First, the PV prediction model and the demand prediction model receive observation data from the environment as input and predict future PV production and electricity demand, respectively. The predicted values are then transformed into latent vectors (Z_{PV} , Z_D) through an embedding process. These two latent vectors are combined and used as part of the RL agent's state. In this framework, the prediction models serve as components of the world model, since their latent vectors are incorporated into the RL agent's state. Such informative representations facilitate the RL agent's training, making it more efficient and stable.

The RL agent is trained using the Deep Deterministic Policy Gradient (DDPG) algorithm. The DDPG algorithm is based on the actor-critic structure and is effective for learning policies in continuous action spaces [47]. The agent learns a policy network that takes the state (S_t)

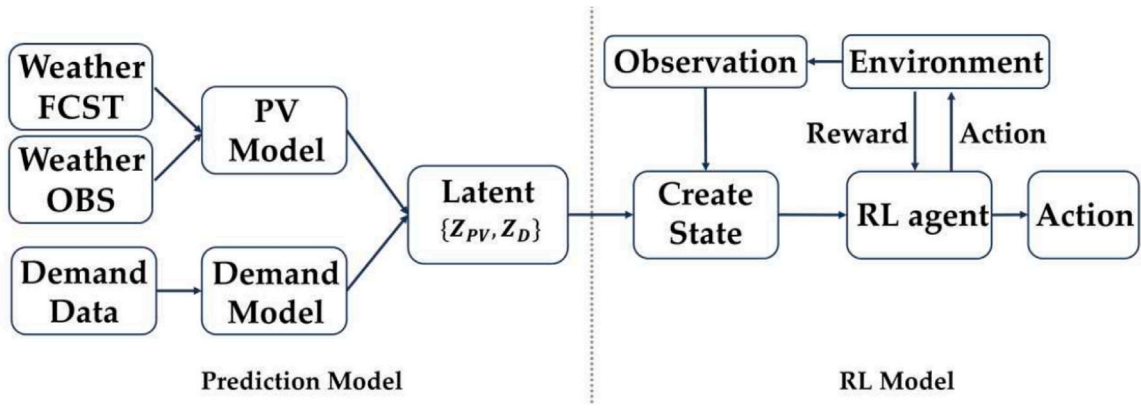


Fig. 6. Proposed flow of World Model-based RL.

as input and produces optimal action (A_t) as output. Here, S_t is formed by combining the information received from the environment (O_t) with the latent information Z_{PV} , Z_D from the prediction models. The World Model-based RL process can be summarized in the following steps:

1. Applying Prediction Models: The prediction models are applied to forecast future information. The predicted values are transformed into latent vectors.
2. State Generation: The latent vectors from the prediction models are combined with the O_t to form an S_t .
3. Action Determination: The actor network takes the S_t as input and outputs the A_t .
4. Transition: The agent interacts with the environment and returns the O_{t+1} and R_t .
5. Policy Update: The critic network updates the parameters to maximize the reward value.

5. Results

This section presents the results of the proposed prediction models and the World Model-based RL approach to train an agent using the DDPG algorithm (WMDDPG). The aim is to answer the following questions through simulations conducted in the SeoulTech campus environment:

1. (Prediction accuracy) Do the proposed PV prediction model and power demand prediction model exhibit sufficient performance?
2. (Prediction horizon) How many future step predictions should be utilized for operation in the World Model-based RL setting?
3. (Cost saving) Does the proposed World Model-based RL operation lead to cost savings?
4. (Intelligent behavior) Does the proposed World Model-based RL operate the system in a smart and efficient manner considering the impact of degradation?

To address these questions, we present our findings in the following subsections. Section 5.1 describes the simulation environment setup. Next, Section 5.2 presents the prediction accuracy of the proposed PV and demand prediction models. Section 5.3 presents the result of the different prediction horizons on the performance. Lastly, Section 5.4 presents the cost-saving performance and behavioral patterns of the proposed RL agent.

5.1. Simulation setup

The simulation environment was set up to closely resemble the actual operating conditions of the SeoulTech campus. Table 4 presents the key specifications of the environment. The ESS capacity was assumed to be 500 kWh, which is consistent with the capacity currently in use at

Table 4

Experiment setting.

Component	Value
Initial ESS capacity	500 kWh
ESS unit price	300,000 KRW/kWh
Initial SoC Level	0%
PV capacity	300 kW
Operation time step	1 h

Table 5

Test cases and periods.

Season	Test case	Train period (30 days)	Test period (10 days)
Winter	Case 1	1/31 - 3/1	3/2 - 3/11
	Case 2	2/10 - 3/11	3/12 - 3/21
	Case 3	2/20 - 3/21	3/22 - 3/31
Summer	Case 4	7/3 - 8/1	8/2 - 8/11
	Case 5	7/13 - 8/11	8/12 - 8/21
	Case 6	7/23 - 8/21	8/22 - 8/31

the university. The ESS unit price was set at 300,000 KRW/kWh, based on the two reports mentioning the range of ESS prices. A report by [48] predicts an average ESS price of around 152 US dollars. Another report shows that ESS prices can vary from 300\$ to 600\$ per kWh [49]. In this study, we approximate the price of ESS as 300,000 Korean Won, which is equivalent to 220 US dollars (1,364 KRW/USD). The PV capacity was set at 300 kW, corresponding to an ESS capacity of about 1.7 h of PV output, which is within the commonly adopted a few hours range that balances economic efficiency and operational benefits [50,51]. The operation was performed on an hourly basis.

To evaluate the performance of the proposed World Model-based RL approach, six test cases were designed using data from February–March and July–August, as shown in Table 5. This captures both winter and summer seasonal patterns, enabling the evaluation under diverse operating conditions. Each test case consists of 30 days of training data and 10 days of testing data. The test cases were divided in this manner to increase data utilization and allow for a more efficient and comprehensive evaluation of the algorithm's performance.

To assess the performance of the trained World Model-based RL agent, three benchmark methods were employed: linear programming (LP) [52], quadratic programming (QP) [53], and Deep-Q-Learning (DQN) RL algorithm [54]. Additionally, a scenario without an ESS (No ESS) was included for comparison. The optimization techniques, QP and LP, utilize accurate information about future power generation and consumption to find the optimal solution. In the PV-ESS micro-grid environment, which exhibits non-linearity, the QP optimization values serve as the ground truth and represent the optimal solution. In contrast, RL agents operate in an uncertain environment where

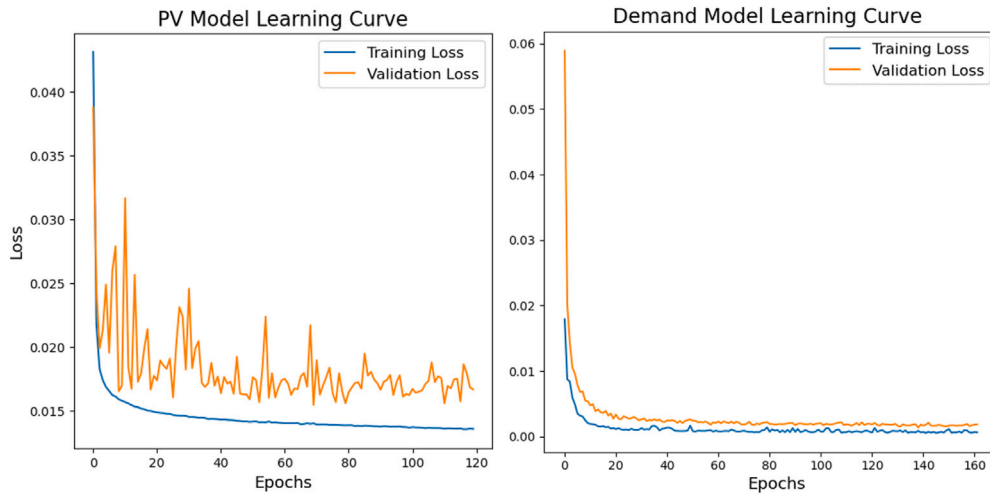


Fig. 7. Learning curve for prediction models.

Table 6
Prediction models accuracy comparison.

Model	PV models		Demand models	
	R^2	RMSE	R^2	RMSE
Our model	0.95	0.04	0.98	0.02
XGBoost	0.93	0.07	0.81	0.06
SARIMA	0.31	0.21	0.47	0.09
Average baseline	0.70	0.14	0.69	0.07

accurate future information is unavailable. They rely on prediction values from previously presented PV generation and demand prediction models to determine the optimal operation strategy. In the No ESS scenario, there is no specific operation method involved; the generated PV energy is immediately consumed at each time step, and the required electricity is purchased from the supplier. The performance of proposed World Model-based RL is evaluated based on its ability to converge to the optimal solution obtained by QP and its superiority over other benchmark methods.

5.2. Prediction model

To address the prediction accuracy from the first question, the performance of the PV prediction model and power demand prediction model was evaluated using a learning curve and appropriate metrics such as root mean square error (RMSE) and the coefficient of determination (R^2). For comparison, we also implemented several baseline models: (1) XGBoost, a widely used ensemble-based decision tree model; (2) SARIMA, a statistical time-series model that captures seasonality and autoregressive patterns; and (3) a simple average baseline, which serves as a naïve benchmark. XGBoost was trained with 500 boosting rounds, a maximum depth of 8, and a learning rate of 0.05. SARIMA ($p, d, q(P, D, Q)_s$ parameters were selected via grid search based on AIC, final configuration of (1, 1, 2)(1, 1, 2)₂₄. The average baseline used the mean of the same weekday over the past three weeks.

Fig. 7 depicts the training and validation loss curves for our two models, with training loss represented in blue and validation loss in orange. Both models demonstrate a decrease in their respective losses as training progresses, indicating a successful learning process.

By incorporating observation data and forecast data, our PV prediction model achieves a high level of accuracy, showing an R^2 score of 0.95 and an RMSE of 0.04. This substantial score attests to the model's exceptional predictive capabilities, underscoring its capacity to deliver precise and reliable forecasts of PV production. Compared to benchmarks, the XGBoost achieved an R^2 of 0.93 with an RMSE of

Table 7
World Model-based RL performance by prediction hour.

Test Case \ Model	1H	3H	5H	12H
Case 1	9101.87	9096.66	9172.94	9189.23
Case 2	7944.81	7943.76	8028.86	8032.45
Case 3	7060.66	7086.89	7135.68	7339.41
Case 4	5591.06	5560.78	5707.61	6008.8
Case 5	5208.86	5196.72	5706.56	5730.86
Case 6	4649.02	4609.83	4757.43	4631.30

0.066, while SARIMA and the average baseline showed substantially weaker performance, $R^2 = 0.307$ and 0.698 , respectively. Additionally, Fig. 8 shows the time-series prediction line graph, which presents the predicted data and target data averaged over 240-hour intervals. This further demonstrates the model's strong predictive power.

Similarly, the power demand prediction model also demonstrates excellent performance, with an R^2 score of 0.98 and an RMSE score of 0.02 consistently outperforming the baselines. XGBoost achieved R^2 of 0.81, RMSE of 0.06, while SARIMA ($R^2 = 0.47$, RMSE = 0.09), and the average baseline ($R^2 = 0.69$, RMSE = 0.07) were unable to capture the complex fluctuations. A comparison of prediction performance across all models for both PV and demand forecasting is summarized in Table 6. Fig. 9 presents a performance evaluation of the demand prediction model with a time-series prediction line graph over 120-hour intervals, further illustrating the model's strong performance despite complex demand fluctuations.

5.3. World model prediction period

In relation to the second question on the number of future horizons to use in the World Model-based RL, experiments were conducted with four different prediction horizons, including 1H, 3H, 5H, and 12H. Table 7 presents the performance of different prediction hours across six test cases. The 1H model directly utilizes the predictions without any embedding, while the 3H and 5H models employ a two-dimensional embedding, and the 12H model uses a five-dimensional embedding. The results show that utilizing short prediction hours yields better performance compared to longer prediction horizons.

In Fig. 10, the average performance across test cases and the score of the prediction models are compared for different prediction hours. It is evident that using predictions from the distant future, such as 12 h ahead, leads to a decrease in performance. This can be attributed to the lower accuracy of long-term predictions, as indicated by the lower scores for 5H and 12H prediction models. Using predictions from the

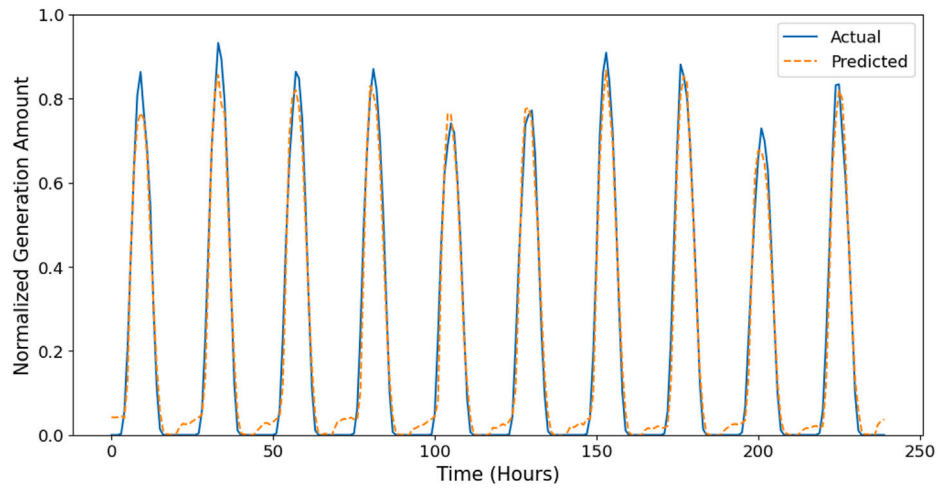


Fig. 8. Actual and predicted PV generation over 240-hours.

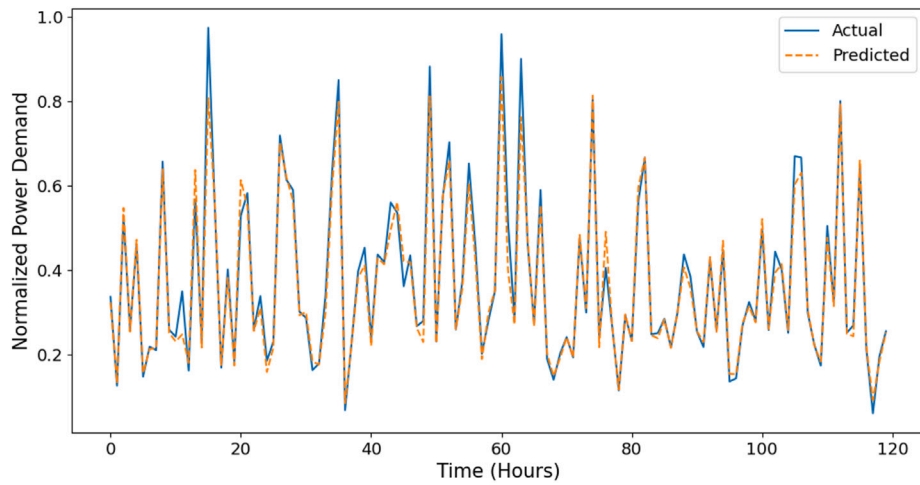


Fig. 9. Actual and predicted power demand over 120-hours.

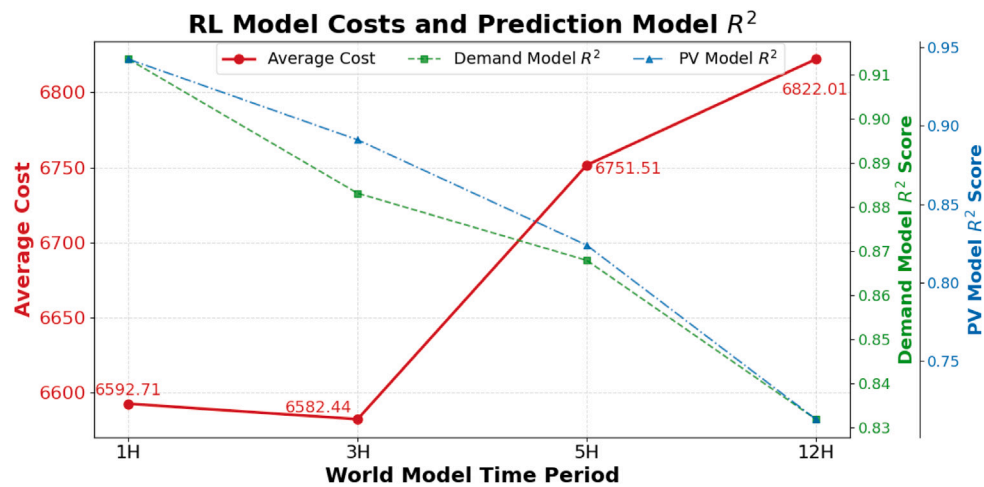


Fig. 10. Operation costs and prediction model accuracy.

near future is more beneficial than relying on distant future predictions. Additionally, the optimal prediction horizon may vary depending on the specific test case, as we can see in the results in Table 7, where the 1H model performs best in one case, while the 3H model is superior in

others. In summary, it is important to select an appropriate prediction horizon for effective operation considering a trade-off: longer horizons include more noisy data but also provide richer information. Based on this consideration, our model adopts the 3H horizon.

Table 8

Cost breakdown and comparison across various models.

Test Case	Model	Power Cost	Deg. Cost	Total Cost	Optimal Gap (%)
1	QP (Optimal)	7824.46	1117.12	8941.59	–
	WMDDPG	8067.84	1028.81	9096.66	1.73
	DQN	8084.39	1041.41	9125.81	2.06
	LP	7632.52	1638.18	9270.70	3.68
	No ESS	10501.61	–	10501.61	17.45
2	QP (Optimal)	6756.19	991.77	7747.96	–
	WMDDPG	6933.98	1009.77	7943.76	2.53
	DQN	6987.92	975.46	7963.39	2.78
	LP	6529.99	1529.94	8059.93	4.03
	No ESS	9905.04	–	9905.04	27.95
3	QP (Optimal)	6056.92	885.53	6942.45	–
	WMDDPG	6037.98	1022.67	7060.66	1.70
	DQN	6083.35	1025.95	7109.31	2.40
	LP	5769.05	1497.21	7266.26	4.66
	No ESS	8192.89	–	8192.89	18.01
4	QP (Optimal)	4275.24	1201.06	5476.3	–
	WMDDPG	4661.86	898.92	5560.78	1.52
	DQN	4968.82	1046.22	6015.04	8.96
	LP	4262.65	1437.42	5700.07	3.93
	No ESS	7154.85	–	7154.85	23.46
5	QP (Optimal)	3507.71	983.86	4491.57	–
	WMDDPG	4401.14	795.58	5196.72	13.57
	DQN	4204.00	1105.83	5309.83	15.41
	LP	3467.97	1195.27	4663.24	3.68
	No ESS	6488.21	–	6488.21	30.77
6	QP (Optimal)	3570.50	1026.28	4596.78	–
	WMDDPG	3654.74	955.09	4609.83	0.28
	DQN	4109.04	1113.96	5223.00	11.99
	LP	3510.94	1283.49	4794.43	4.12
	No ESS	6017.37	–	6017.37	23.61

5.4. Main result

The main result demonstrates the trained agent's ability to save costs and intelligent behavior in various test cases. The comparison results between the proposed WMDDPG and other benchmark models are presented in Table 8, addressing the third question. The table shows the breakdown of power cost, degradation cost, and total cost for each model across three test cases. The QP model, which serves as the ideal solution, is highlighted in blue. The QP model is considered an ideal solution because it operates based on perfect future information rather than relying on predictions.

The WMDDPG model achieves near-optimal performance in the majority of test cases. In the winter scenarios (Cases 1–3), with Test case 1, WMDDPG incurs a total cost of 9096.66, which is only 1.73% higher than the optimal result obtained from QP. Similarly, in Test Cases 2 and 3, WMDDPG exhibits a small difference of 2.53% and 1.70%, respectively, compared to the optimal outcome. In the summer scenarios (Cases 4–6), WMDDPG performs similarly well: Case 4 shows only a 1.52% of optimality gap, and Case 6 nearly matches the QP solution with just a 0.28% optimality gap. While Case 5 shows a somewhat larger deviation, WMDDPG outperforms most of the benchmark models and remains close to the optimal solution under varying seasonal conditions.

In contrast, the other benchmark models demonstrate lower cost savings compared to WMDDPG. The DQN model incurs total costs that are 0.33% to 11.71% higher than the WMDDPG model across the six test cases. The LP model, despite having lower power costs, incurs significantly higher degradation costs, resulting in total costs that are 1.91% to 3.84% higher than the WMDDPG model. This result comes from the non-linear nature of battery degradation, which cannot be effectively represented with the linear structure of the LP model. Therefore, this leads to suboptimal performance in terms of degradation costs despite achieving greater electricity cost savings. These results

indicate that the proposed WMDDPG model comes closest to the optimal solution and provides better overall cost savings than the other benchmark models.

To address the last question, Fig. 11 illustrates the hourly power purchase and battery usage patterns. The bar chart represents the power purchased from the supplier, while the line chart indicates the battery power flow, with positive values denoting charging and negative values representing discharging.

The proposed WMDDPG model demonstrates intelligent decision-making by minimizing power purchases during peak hours and strategically charging the battery in the hours leading up to the peak period. With battery degradation considerations, the model consistently charges 2–3 h before the peak in both summer and winter. Specifically, charging the battery too early and leaving it over a long duration may cause SoC stress, while rapid charging, for example charging all at once right before peak hour, causes significant DoD stress. By balancing these two stress factors, our agent adopts a more gradual charging approach that minimizes both SoC and DoD stress. These observations highlight the effectiveness of the WMDDPG model in intelligently managing power purchases and battery usage to minimize costs while considering battery degradation factors, demonstrating its potential for achieving cost savings and efficient energy management in PV-ESS operation.

Unlike mathematical optimizations, such as LP or QP, which assume perfect information, or MPC, which applies a fixed prediction horizon and depends heavily on forecast accuracy, the WMDDPG adaptively learns how to exploit predictions. By embedding forecasts into latent features and adjusting their importance through feedback, the agent can remain robust to prediction errors and nonlinear degradation effects. This adaptive approach further strengthens the inherent advantage of RL frameworks in handling uncertainty, compared to conventional methods that treat all forecasts equally.

6. Conclusion

In this study, we proposed an innovative approach to optimize the operation of PV-ESS systems using a World Model-based RL. Our approach addresses the limitations of traditional mathematical optimization methods by effectively handling uncertainties. By integrating advanced prediction models for PV generation and power demand, as well as a comprehensive degradation model for ESS, we have demonstrated significant improvements in cost reduction and operational efficiency. To summarize the key findings based on our research questions:

- Performance of Prediction Models:** The proposed PV and power demand prediction models exhibit high predictive performance, achieving R^2 scores of 0.95 and 0.98 - outperforming baseline models such as XGBoost, SARIMA, and simple averaging.
- Efficiency of Future Step Predictions:** Utilizing short-term predictions of 3 h is efficient in the World Model-based RL setting considering the trade-off between decreasing accuracy and the benefit of utilizing future information.
- Cost Savings:** The proposed World Model-based RL operation model achieves significant cost savings, particularly in environments without batteries, reducing costs by 15.7–27.1%.
- Smart Operation:** The model operates the system efficiently, considering ToU pricing and battery degradation.

Future work should focus on further enhancing the prediction models and exploring the application of this framework to other renewable energy systems. We plan to use more extensive datasets to compare seasonal differences and apply our algorithm to various microgrid environments to evaluate its effectiveness. The insights gained from this study provide a strong foundation for the development of more resilient

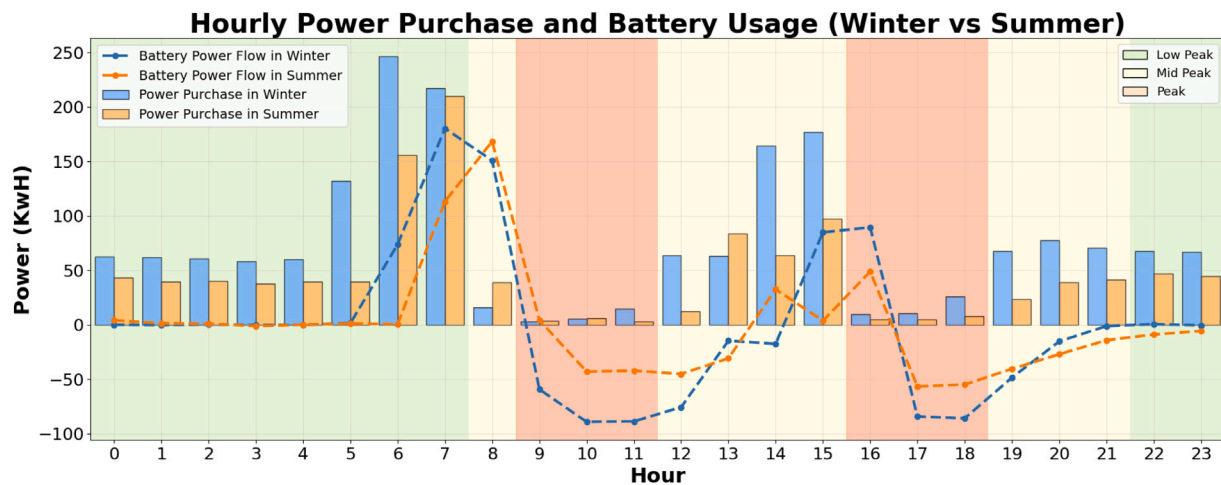


Fig. 11. Hourly power purchase and battery usage.

and efficient energy management strategies in response to growing environmental and economic challenges.

CRedit authorship contribution statement

Jaemin Park: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Taehyeon Kwon:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Min K. Sim:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Min K. Sim reports financial support was provided by National Research Foundation of Korea. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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